

ONLINE QUIZ APPLICATION SYSTEM

AI23431 – WEB TECHNOLOGY AND MOBILE APPLICATION

MINI-PROJECT REPORT

**SUBMITTED BY
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in partial fulfillment for the award of the degree of

**BACHELOR OF TECHNOLOGY
in
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE
RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS),
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MAY 2026

RAJALAKSHMI ENGINEERING COLLEGE
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ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. M. Gnanasekar M.E., Ph.D.**, Professor and Head of the Department of Artificial Intelligence and Data Science for his guidance and encouragement throughout the project work.

We convey our sincere and deepest gratitude to our internal guide, **Mr. G. Thiagarajan**, Department of Artificial Intelligence and Data Science, for his useful tips during our review to build our project. The Online Quiz Application System presented in this report is a result of his invaluable mentorship and the supportive academic environment at Rajalakshmi Engineering College.

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ABSTRACT

The Online Quiz Application System (OQAS) presents a comprehensive digital platform for conducting, managing, and evaluating subject-based quizzes within educational institutions. With the increasing demand for interactive and self-paced learning tools, the system is designed to replace traditional paper-based assessments with a secure, automated, and scalable web application. The platform enables students to register and log in using OTP-based email or phone authentication, ensuring secure access, while providing a rich interface to attempt subject-wise unit quizzes and track their academic progress in real time.

The application is structured around five subjects — Mathematics, Science, Computer Science, English, and General Knowledge — each divided into five units with fifteen multiple-choice questions per unit, totaling 375 questions. A built-in countdown timer of twenty minutes is enforced per unit attempt to simulate real examination conditions. Upon completion of each unit quiz, students receive an instant result breakdown including their score, percentage, grade classification, and a detailed question-by-question review highlighting correct and incorrect responses.

The architecture of the system employs a three-tier client-server design consisting of a responsive single-page HTML/CSS/JavaScript frontend, a Node.js Express backend, and a lightweight lowdb JSON-file database. Role-based JWT authentication and bcrypt password hashing ensure that data remains secure and access is limited to verified users. An OTP verification workflow — supporting both email via SMTP Nodemailer and SMS via Twilio — guards the registration and login process, preventing unauthorized access.

The dashboard module presents a comprehensive performance overview showing subject-level progress, unit-wise completion bars, average scores, and best scores across all attempted quizzes. A result history table records all past attempts with timestamps and time taken, enabling long-term academic monitoring. Overall, the Online Quiz Application System improves assessment accuracy, reduces administrative workload, enhances student engagement, and provides reliable analytics for continuous academic improvement.

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CHAPTER 1: INTRODUCTION

1.1 General

Online assessment and quiz systems play an increasingly vital role in modern education as they provide an interactive, automated, and scalable way to evaluate student knowledge. Traditional paper-based examinations and manual quiz systems are time-consuming, resource-intensive, and difficult to scale for large batches of students. With the advancement of web technologies and cloud infrastructure, digital quiz platforms have emerged as an efficient alternative that enables real-time evaluation, instant feedback, and centralized performance tracking.

The Online Quiz Application System (OQAS) is designed to automate the entire quiz lifecycle — from user registration and OTP verification, to quiz attempt, automated grading, and performance analytics. The system provides students with an interactive interface to attempt subject-wise unit quizzes, review their answers, and monitor their academic progress through a personalized dashboard. By leveraging modern web technologies including Node.js, Express.js, and a JSON-based database, the system operates efficiently on standard local hardware without requiring complex infrastructure.

1.2 Need for Online Quiz Application System

Many educational institutions still rely on conventional examination methods that involve preparing question papers, manual invigilation, and manual evaluation. These processes are prone to errors, require significant faculty time, and do not provide students with immediate feedback. Furthermore, the absence of a centralized digital record makes it difficult to track individual student progress over time.

A digital quiz application addresses these challenges by automating question delivery, response capture, evaluation, and result generation. Students can attempt quizzes at their convenience, receive instant scores, and review their performance in detail. Faculty and administrators benefit from centralized progress data that facilitates timely intervention for students struggling in specific units or subjects.

The OQAS specifically addresses the need for a lightweight, browser-based quiz system that requires no additional software installation, supports OTP-secured registration and login, and provides a rich analytical dashboard — all within a single integrated application.

1.3 Scope of the System

The Online Quiz Application System is developed to provide a comprehensive digital platform for subject-wise unit-based quizzes accessible through any modern web browser. The system supports five core academic subjects — Mathematics, Science, Computer Science, English, and General Knowledge — each containing five units with fifteen multiple-choice questions per unit, totaling 375 questions across the platform.

The scope includes secure user registration with OTP verification via email or SMS, JWT-based session management, a timed quiz engine with a twenty-minute countdown per unit, automatic grading and percentage calculation, a detailed result review showing correct and incorrect answers, a personal dashboard tracking progress across all subjects and units, and a complete result history for repeated attempts.

The system can be deployed in schools, colleges, coaching institutions, and online learning environments without specialized hardware requirements. It is built with extensibility in mind, allowing additional subjects, units, and question types to be integrated in future versions.

1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the Online Quiz Application System:

- Chapter 1 provides the introduction, including general overview, need for the system, and scope of the project.
- Chapter 2 presents the literature survey discussing existing quiz and assessment systems and related research work.
- Chapter 3 explains the problem formulation and specific objectives of the proposed system.
- Chapter 4 describes the system design including architecture, database design, and hardware/software requirements.
- Chapter 5 discusses the implementation of each system module in detail with supporting screenshots.
- Chapter 6 presents the results, performance evaluation, and system testing outcomes.
- Chapter 7 concludes the report and discusses possible future enhancements.

1.5 System Architecture Diagram

The system follows a three-tier web architecture as illustrated below:

Presentation Layer	HTML + CSS + JavaScript (Single Page Application)
Application Layer	Node.js + Express.js (REST API Backend) · JWT Authentication · OTP via Nodemailer / Twilio
Data Layer	lowdb (JSON file database) · Users · OTPs · Quiz Results
Question Bank	5 Subjects × 5 Units × 15 MCQs = 375 Questions (questions.js)

CHAPTER 2: LITERATURE SURVEY

2.1 Introduction

Online quiz and assessment systems have become increasingly prevalent in educational technology, with numerous research studies and commercial implementations demonstrating their effectiveness over traditional manual methods. The proliferation of web-based technologies including JavaScript frameworks, cloud databases, and secure API architectures has made it feasible to develop lightweight yet powerful quiz platforms that operate entirely through a web browser. This literature survey examines existing quiz and assessment systems, identifies their strengths and limitations, and contextualizes the design decisions made in the development of the Online Quiz Application System.

2.2 Existing Quiz and Assessment Systems

Several established platforms for online assessment exist in the academic and commercial space. Learning Management Systems (LMS) such as Moodle, Canvas, and Google Classroom incorporate quiz modules that allow educators to create, distribute, and auto-grade assessments. While these platforms are comprehensive, they require significant infrastructure setup, administrative overhead, and often depend on internet connectivity to cloud servers, making them unsuitable for lightweight local deployment.

Simpler browser-based quiz tools such as Kahoot, Quizizz, and Typeform offer gamified quiz experiences but lack structured academic features such as unit-wise progression, subject-based organization, historical performance tracking, and detailed per-question answer review. These tools are primarily designed for classroom engagement rather than formal assessment and progress monitoring.

Research studies have explored various technologies for automated assessment including RFID-based attendance and quiz systems, mobile application-based quizzes, and AI-driven adaptive testing. These approaches, while innovative, require specialized hardware or complex machine learning infrastructure that is not accessible in standard educational environments.

2.3 Related Work

Delabaty and Al-Turaiki (2022) conducted a systematic literature review of automated assessment management systems and found that web-based systems significantly outperform manual methods in terms of accuracy, efficiency, and student satisfaction. Their review highlighted the importance of real-time feedback,

centralized data management, and role-based access control as critical features for academic assessment platforms.

Studies focusing on Node.js for educational backend systems (Node.js Foundation Documentation) demonstrate that JavaScript-based server environments provide excellent performance for concurrent user handling, making them well-suited for quiz applications where multiple students may submit answers simultaneously. The event-driven, non-blocking I/O model of Node.js ensures low latency responses even under moderate server load.

Research into OTP-based authentication for educational portals (Zafar et al., 2015; related works on SMTP and two-factor authentication) emphasizes that verification workflows combining password authentication with one-time passwords significantly reduce unauthorized access incidents. Implementing both email and SMS OTP channels, as done in the OQAS, provides flexibility while maintaining security standards.

2.4 Summary

From the literature survey, it is evident that automated quiz systems provide significant advantages over traditional assessment methods including improved accuracy, instant evaluation, and centralized data management. Existing comprehensive platforms are often over-engineered for small-scale institutional use, while simpler tools lack the structured academic features required for formal assessment. The Online Quiz Application System fills this gap by providing a purpose-built, lightweight, browser-based quiz platform with OTP security, subject-unit organization, a timed quiz engine, detailed result analytics, and a progress dashboard — all accessible without additional software installation.

CHAPTER 3: PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

Many educational institutions continue to rely on traditional assessment methods involving printed question papers, manual invigilation, and manual evaluation. These conventional approaches present several critical challenges that hinder efficient academic monitoring and student engagement:

- Manual question paper preparation is time-consuming and susceptible to errors, reprinting costs, and security breaches through paper leakage.
- Students receive evaluation results days or weeks after an assessment, depriving them of immediate feedback necessary for timely course correction.
- Faculty members spend significant time evaluating scripts and computing scores manually, reducing the time available for teaching and curriculum development.
- There is no centralized digital record of student performance across multiple subjects and units, making it difficult to identify struggling students and track academic progress over time.
- Existing commercial quiz platforms require internet access to cloud servers, licensing fees, and complex administrative setup, making them inaccessible for many institutions.

Therefore, a need exists for a lightweight, self-hosted, browser-based quiz system that automates question delivery, response evaluation, and performance analytics — while remaining accessible on standard hardware without specialized infrastructure or recurring costs.

3.2 Objectives of the System

The main objective of the Online Quiz Application System is to develop a full-stack digital platform that automates subject-based unit quiz management with secure authentication, instant evaluation, and detailed performance analytics. The specific objectives include:

- To implement OTP-based registration and login using email or phone number, ensuring secure access for all users.
- To organize 375 multiple-choice questions across 5 subjects, 5 units per subject, and 15 questions per unit for structured academic assessment.

- To enforce a 20-minute countdown timer per quiz attempt to simulate real examination conditions.
- To provide instant automated evaluation with score, percentage, grade classification, and per-question answer review upon quiz completion.
- To maintain a comprehensive personal dashboard showing subject-level progress, unit completion status, average scores, and best scores.
- To store all quiz result history in a persistent database enabling long-term academic performance tracking.
- To operate as a zero-installation, browser-based single-page application deployable on any standard local machine with Node.js installed.

CHAPTER 4: SYSTEM DESIGN

4.1 Introduction

System design is a critical phase in software engineering where the overall structure, component interactions, data flows, and technical specifications of the system are defined. A well-structured design ensures that the system is efficient, maintainable, and scalable. The Online Quiz Application System adopts a modular three-tier architecture that separates the user interface, business logic, and data storage into distinct layers, enabling clean component boundaries and independent development of each module.

4.2 System Architecture

The system follows a client-server architecture consisting of three primary layers:

Presentation Layer: A single-page application (SPA) built with HTML5, CSS3, and vanilla JavaScript. All pages — authentication, subject browser, quiz engine, result display, and dashboard — are served from a single `index.html` file with dynamic rendering managed by client-side JavaScript. The interface is responsive and accessible on desktop and mobile browsers.

Application Layer: The backend is implemented using Node.js with the Express.js framework. It handles REST API routes for authentication (register, send-OTP, verify-OTP, login), quiz operations (fetch subjects, fetch units, fetch questions, submit answers), and result retrieval. JWT tokens are used for session management, and bcrypt is used for password hashing.

Data Layer: The database is implemented using lowdb, a lightweight JSON file-based database that requires no native compilation or external database server. It stores user accounts, OTP records, and quiz results. The 375-question bank is maintained as a static JavaScript module (`questions.js`) loaded at runtime.

4.3 Database Design

The system maintains three primary data collections in the JSON database:

Collection	Fields	Description
users	id, name, email, phone, password (hashed), is_verified, created_at	Stores registered user accounts with verification status

Collection	Fields	Description
otps	id, identifier, otp, expires_at, used, created_at	Tracks OTP records per identifier with expiry and usage status
results	id, user_id, subject_id, unit_id, score, total, time_taken, completed_at	Records quiz attempt outcomes for each user per unit

4.4 System Requirements

4.4.1 Hardware Requirements

The Online Quiz Application System is designed to operate on standard computing hardware:

Component	Specification
Processor	Intel Core i3 or equivalent (or higher)
RAM	Minimum 4 GB
Storage	Minimum 500 MB free disk space
Network	Local network or internet connection (for email/SMS OTP)
Input Devices	Keyboard and Mouse
Display	Any monitor with 1024×768 resolution or higher

4.4.2 Software Requirements

The following software tools and technologies are required to run the application:

Component	Specification
Operating System	Windows 10 / macOS / Ubuntu 20.04 or later
Runtime Environment	Node.js v16 or higher

Component	Specification
Frontend Technologies	HTML5, CSS3, Vanilla JavaScript (ES6+)
Backend Framework	Express.js 4.x
Database	lowdb 1.0.0 (JSON file-based, zero installation)
Authentication	JSON Web Token (JWT) + bcryptjs
OTP – Email	Nodemailer with Gmail SMTP (App Password)
OTP – SMS	Twilio SDK (optional)
Development Environment	Visual Studio Code
Web Browser	Google Chrome, Firefox, or any modern browser
Package Manager	npm (Node Package Manager)

CHAPTER 5: SYSTEM IMPLEMENTATION

5.1 System Methodology

The Online Quiz Application System is implemented using a modular web-based approach that enables users to access all functionality through a standard browser without additional software installation. The development process follows an iterative methodology where each module was designed, implemented, and tested independently before integration into the full system.

The frontend of the system is developed as a single HTML file containing all CSS styles and JavaScript logic, implementing a single-page application (SPA) paradigm where different views are dynamically rendered by toggling CSS display states. This approach eliminates page reloads and provides a seamless user experience. API communication is handled using the Fetch API with JSON payloads and Bearer token authorization headers.

The backend follows a RESTful API design where Express.js routes handle distinct functional domains — authentication and quiz operations — each isolated in separate route modules. The middleware layer enforces JWT authentication on all protected endpoints, ensuring that only verified users can access quiz content and result data. The questions module is a standalone JavaScript data file that serves as a static embedded database requiring no SQL queries or ORM overhead.

5.2 Module Descriptions

The Online Quiz Application System consists of six primary modules that work together to deliver a seamless quiz experience:

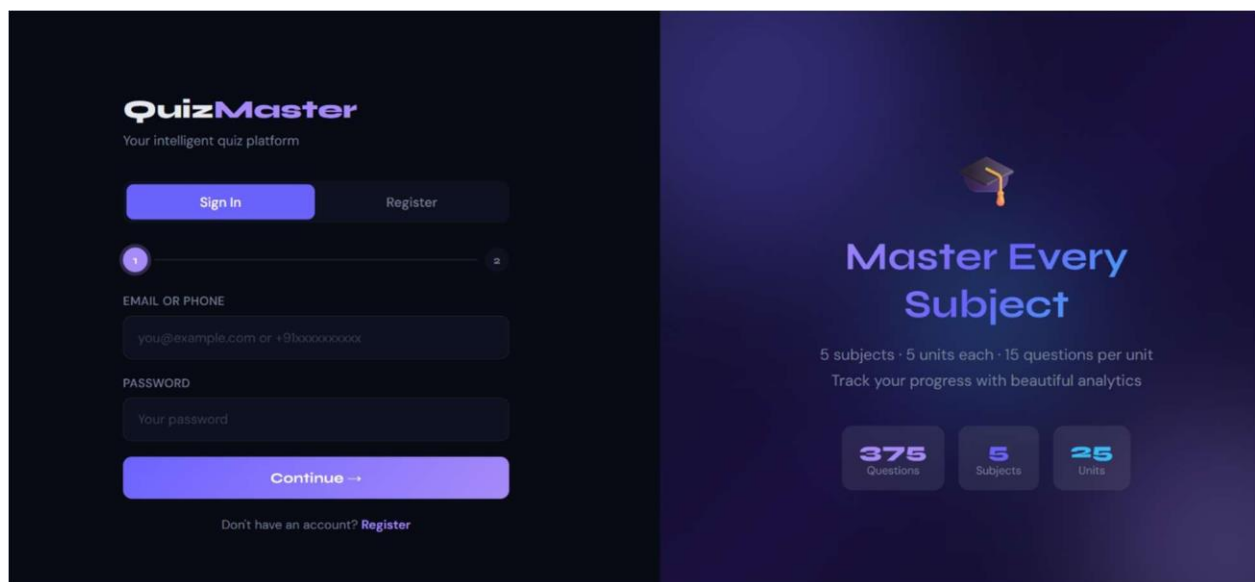
5.2.1 Authentication Module

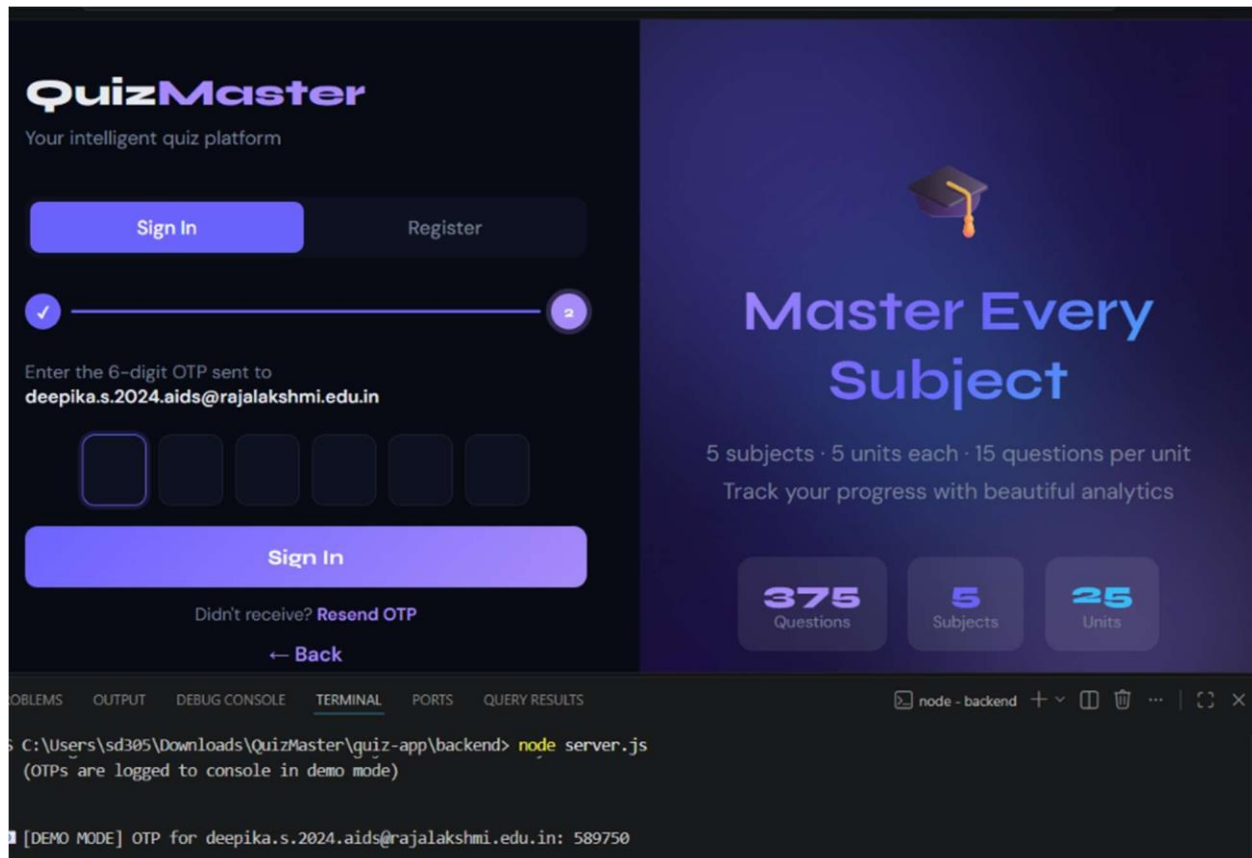
The Authentication Module manages the complete user lifecycle from registration through verified login. It implements a multi-step OTP verification flow to ensure that only legitimate users with confirmed email or phone access can log into the system.

Registration Flow: The user provides their full name, email or phone number, and a password (minimum 6 characters). The backend validates the input, checks for duplicate registrations, hashes the password using bcrypt with a salt factor of 10, creates the user record in the database with verification status set to false, and sends a 6-digit OTP to the provided identifier via email (Nodemailer/SMTP) or SMS (Twilio). The user enters the OTP in the browser, and upon successful verification, the account is marked as verified.

Login Flow: Registered and verified users enter their email or phone number and password. The backend locates the user record, verifies the password using bcrypt comparison, sends a new OTP to the identifier, and waits for OTP confirmation before generating a signed JWT token with a 7-day expiry. The token is stored in the browser's localStorage and attached as a Bearer header on all subsequent API requests.

Security Measures: OTPs are invalidated immediately upon use and expire after 10 minutes (configurable). JWT tokens are signed with a server-side secret key. All passwords are stored as bcrypt hashes and never transmitted in plain text. The OTP input uses a six-box split input interface for improved UX and reduced transcription errors.



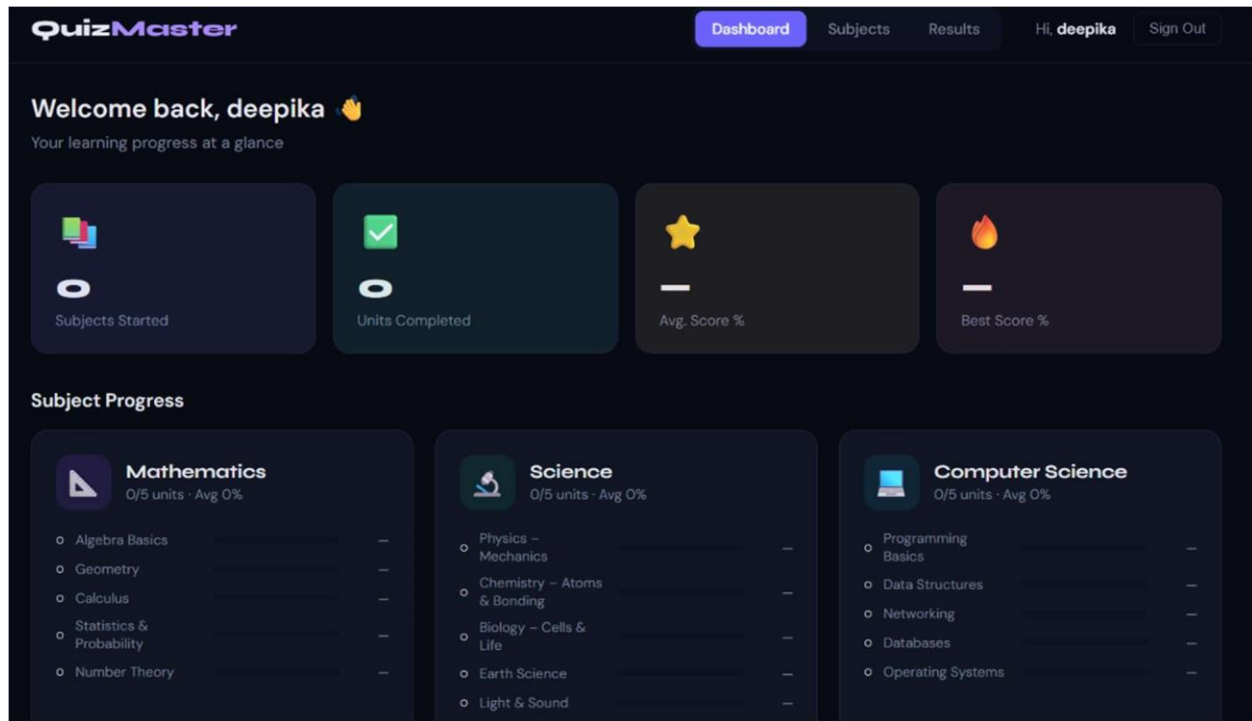


5.2.2 Subject and Unit Browser Module

The Subject and Unit Browser Module provides the navigational structure for the quiz application, allowing students to explore available subjects and select specific units for assessment.

Subject Browser: Upon login, the subjects tab displays all five available subjects — Mathematics, Science, Computer Science, English, and General Knowledge — as styled cards, each showing the subject icon, description, and number of units completed by the logged-in user. The cards are color-coded by subject using distinct accent colors to enhance visual differentiation.

Unit Browser: Selecting a subject transitions to the unit view, which lists all five units for that subject. Each unit card displays the unit name, number of questions, estimated time, and — if previously attempted — the score achieved with a color-coded badge (green for $\geq 80\%$, amber for 50–79%, red for below 50%). Completed units show a distinct left border to communicate completion status at a glance. Users can retake any unit regardless of prior completion.



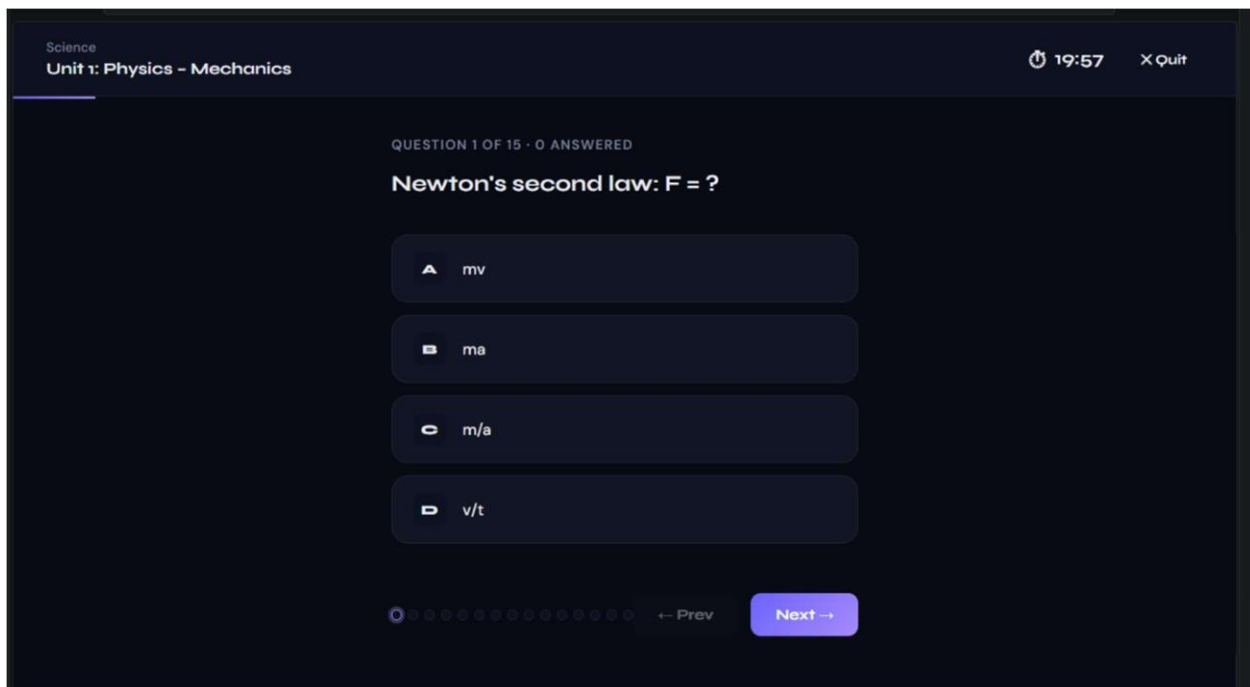
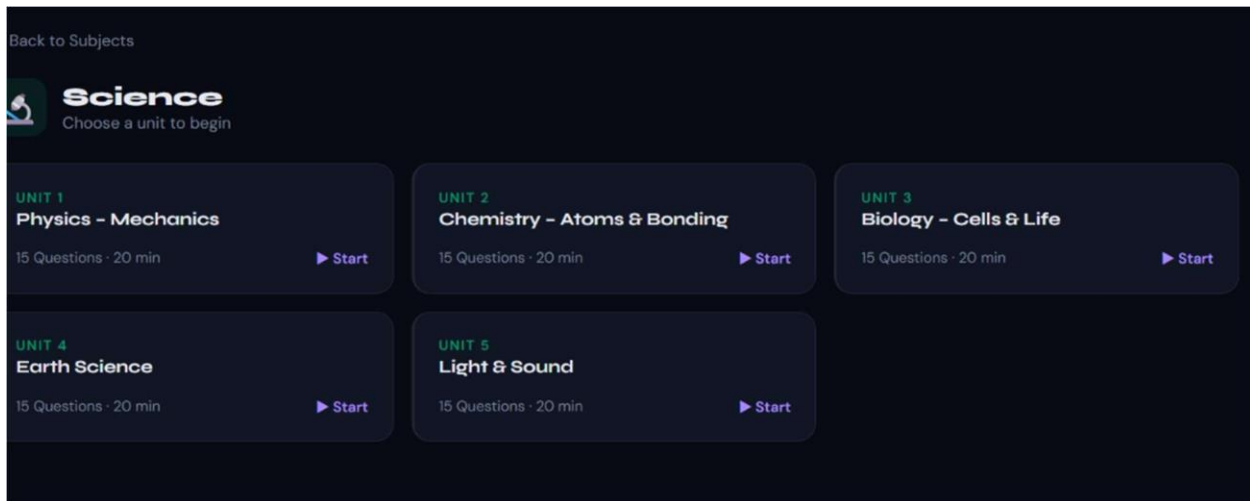
5.2.3 Quiz Engine Module

The Quiz Engine Module is the core component of the system, handling question delivery, response capture, timer management, and navigation throughout the quiz session.

On initiating a quiz, the backend API returns the 15 questions for the selected unit without including the correct answer indices, preventing client-side answer inspection. The frontend renders one question at a time with four labeled option buttons (A, B, C, D). Selecting an option highlights it with an accent border and stores the choice in the answers array indexed by question number.

Timer: A 20-minute countdown timer is displayed in the header. The timer changes color from neutral to amber (≤ 3 minutes remaining) and then to flashing red (≤ 1 minute remaining). Upon timer expiry, the quiz is automatically submitted. The time taken is recorded in seconds and sent to the backend with the submission payload.

Navigation: Students can navigate freely between questions using Prev/Next buttons or by clicking the dot indicators at the bottom. Answered dots are highlighted in accent color, and the current question dot has a glowing ring. The Submit button replaces the Next button on the final question. A confirmation dialog is shown if the user attempts to quit mid-quiz.



5.2.4 Result and Analytics Module

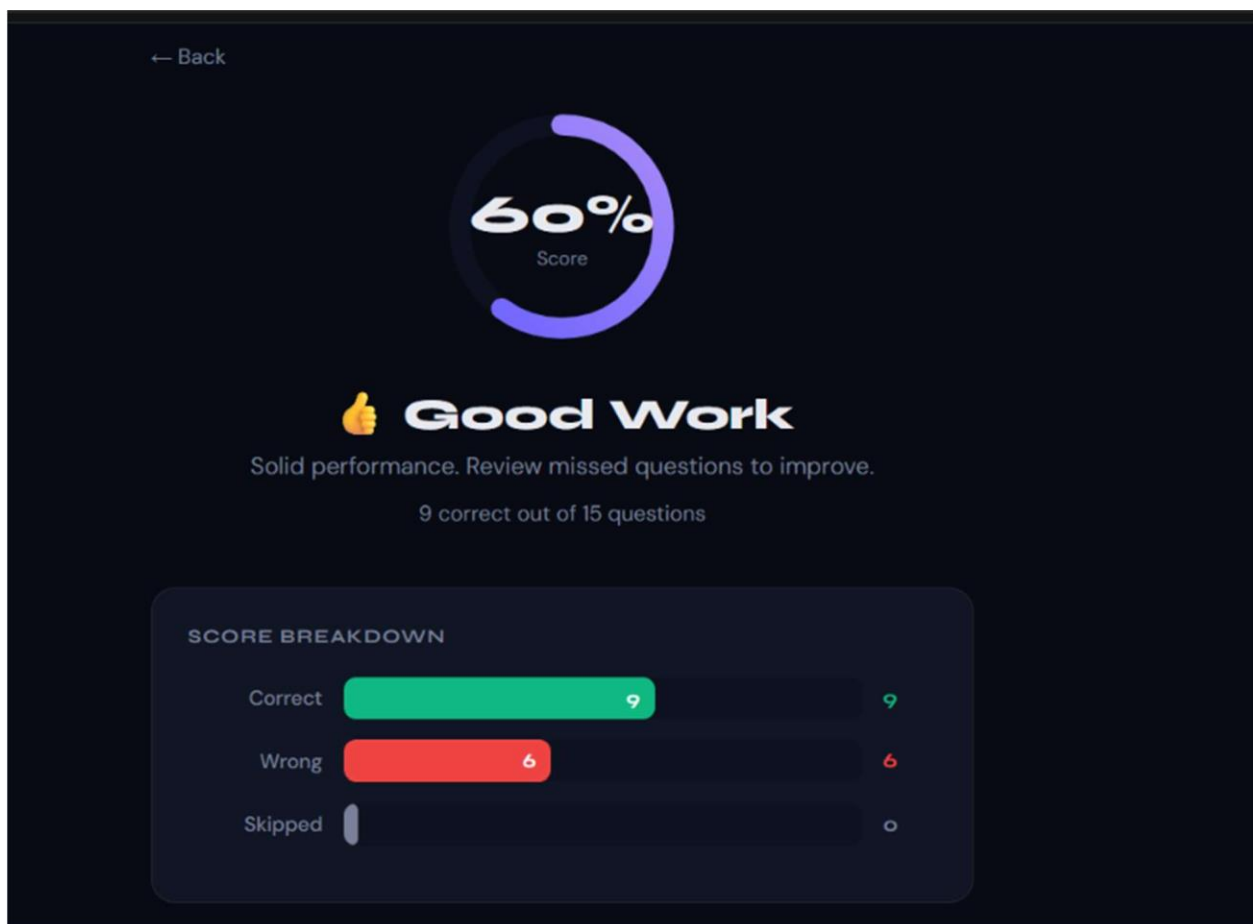
The Result and Analytics Module processes the submitted answer array, calculates the score, and presents a comprehensive result breakdown to the student immediately upon quiz submission.

Score Calculation: The backend compares each submitted answer index against the stored correct answer index for each question. Correct answers increment the score counter. The final score, total questions, percentage, and per-question detail

(question text, all options, user's answer, correct answer, and correctness flag) are returned in the API response.

Result Display: The result page displays an animated circular score ring that fills proportionally to the percentage achieved, showing the score percentage at the center. A grade badge is shown below with five classification levels: Excellent ($\geq 90\%$), Great Job ($\geq 75\%$), Good Work ($\geq 60\%$), Keep Studying ($\geq 40\%$), and Don't Give Up ($< 40\%$). The raw score (e.g., '11 correct out of 15') is shown beneath the grade.

Score Breakdown Bar Chart: Three horizontal bars display the count of Correct, Wrong, and Skipped answers with proportional widths, providing a quick visual summary of performance. A detailed question review section lists all 15 questions with the student's selected answer highlighted in green (correct) or red (incorrect), and the correct answer displayed alongside incorrect responses. Students scoring 75% or above are rewarded with an animated confetti effect.



5.2.5 Dashboard Module

The Dashboard Module provides a centralized performance overview that aggregates all quiz results across subjects and units, enabling students to monitor their academic progress holistically.

Statistics Cards: Four summary cards at the top of the dashboard display: Subjects Started (number of subjects with at least one attempted unit), Units Completed (total number of units attempted), Average Score Percentage (mean across all attempted units), and Best Score Percentage (highest single unit score achieved).

Subject Progress Cards: One card per subject displays the subject icon, name, average score, and a vertical list of all five units. Each unit row shows a progress bar filled proportionally to the score achieved (green for passing, amber for borderline), the exact percentage, and a completion icon. Clicking a subject card navigates directly to that subject's unit browser.

Result History Table: The Results tab presents a paginated table of all past quiz attempts in reverse chronological order, showing subject, unit, score pill (colored by performance tier), time taken, and date of attempt. This provides a comprehensive audit trail for tracking improvement over repeated attempts.

5.2.6 OTP Notification Module

The OTP Notification Module handles the generation and delivery of one-time passwords to users during registration and login. It supports two delivery channels — email via SMTP and SMS via Twilio — configurable through the backend `config.js` file.

OTP Generation: A cryptographically random 6-digit numeric OTP is generated using `Math.random` seeded with system entropy. The OTP is stored in the database with a timestamp and configurable expiry (default 10 minutes). Prior active OTPs for the same identifier are invalidated before a new one is created, preventing OTP enumeration attacks.

Email Delivery: When email mode is enabled, the OTP is sent via the Nodemailer library using a Gmail SMTP connection authenticated with a Gmail App Password. The email template uses HTML styling with the brand accent color and clearly displays the 6-digit code, expiry notice, and a do-not-share warning.

Demo Mode: When email and SMS are not configured (default state), OTPs are logged to the server console with a clear DEMO MODE label, allowing developers and evaluators to test the full authentication flow without configuring external

services. This approach ensures the system remains functional for local evaluation without any external dependency.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The Online Quiz Application System was successfully developed and tested using a web-based platform running locally on Node.js. The system performs all core functions correctly — including user registration with OTP verification, secure JWT-based login, structured subject and unit navigation, a timed quiz engine with 15 MCQs per unit, automatic score calculation, detailed result review, a progress dashboard, and a complete result history table.

During testing, all 375 questions across 5 subjects and 25 units were verified to load correctly in quiz mode without exposing correct answer indices in the API response. The timer countdown, auto-submit functionality, and navigation between questions operated without errors. The result page animations (score ring, confetti for high scores) rendered correctly across tested browsers including Chrome and Firefox.

The OTP delivery system was verified in demo mode (console logging) and in email mode using a configured Gmail SMTP connection. OTPs were generated, stored, validated, and expired as specified. JWT token authentication was verified to block unauthorized API requests with a 401 response when no token or an invalid token was provided.

6.2 Performance Evaluation

Usability: The single-page application design ensures zero page reload delays. Navigation between subjects, units, and quiz sessions is instantaneous. The dashboard renders all subject and unit progress data within a single API call, minimizing network overhead.

Accuracy: Score calculation was verified across multiple test attempts with known correct-answer sets, confirming 100% calculation accuracy. The OTP expiry and invalidation logic was tested for edge cases including reuse attempts and expired OTP submission.

Resource Utilization: The system operates with a Node.js backend consuming under 50 MB of RAM at idle. The lowdb JSON database requires no external server process. The entire application (excluding node_modules) is under 250 KB, making it highly portable and resource-efficient.

6.3 System Testing

Comprehensive test cases were executed to validate all functional requirements of the system. The following table summarizes the test case results:

CHAPTER 7: CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Quiz Application System was successfully designed, developed, and tested as a complete full-stack web application for subject-based unit quiz management. The system replaces traditional paper-based assessments with a secure, automated, and interactive digital platform that is accessible through any modern browser without additional software installation.

The application delivers on all its stated objectives: OTP-secured registration and login via email or phone, a structured quiz framework covering 375 questions across 5 subjects and 25 units, a timed quiz engine with automatic submission and instant grading, detailed result analytics with a score ring visualization and per-question review, and a comprehensive dashboard tracking progress across all subjects and units. The modular backend design, RESTful API architecture, and JWT authentication model ensure that the system is secure, maintainable, and extensible.

The project successfully demonstrates the integration of modern web technologies — Node.js, Express.js, lowdb, JWT, bcrypt, and Nodemailer — into a cohesive academic quiz platform. It provides a practical foundation for further development towards a production-grade educational assessment system.

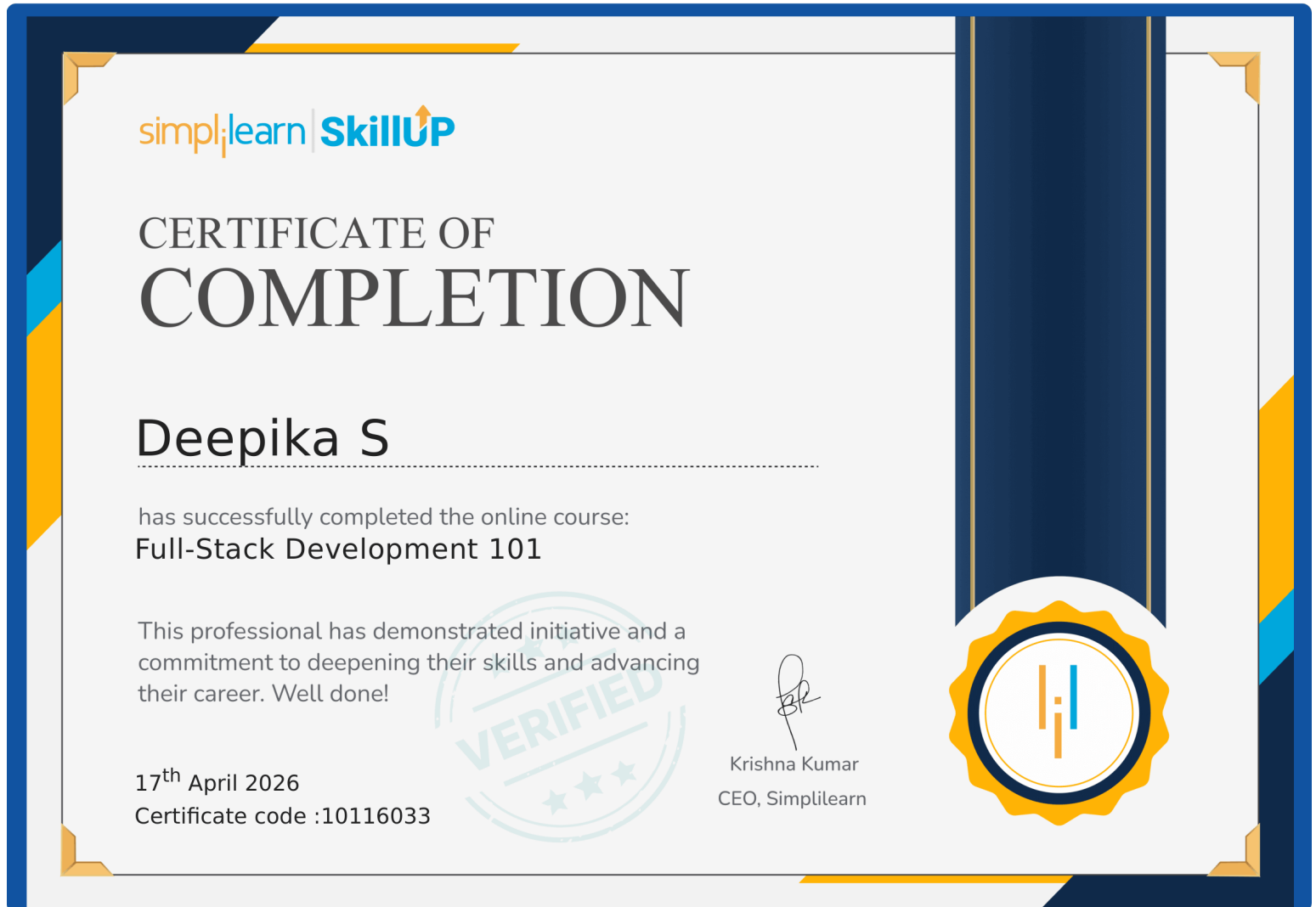
7.2 Future Enhancements

Although the current system provides comprehensive features for online quiz management, several enhancements can be made in future iterations to further improve its functionality and scalability:

- **Faculty Portal:** Develop a separate faculty dashboard allowing educators to add, edit, and manage questions dynamically through a web interface, eliminating the need to modify the questions.js file directly.
- **Adaptive Testing:** Integrate machine learning algorithms to adjust question difficulty dynamically based on a student's performance history, delivering personalized assessments.
- **Mobile Application:** Develop native iOS and Android companion apps using React Native or Flutter for a dedicated mobile quiz experience with push notification support.

- **MySQL / PostgreSQL Integration:** Replace the JSON file database with a production-grade relational database server for improved concurrency, transaction support, and scalability to large user bases.
- **Leaderboard and Gamification:** Introduce subject-wise and overall leaderboards, achievement badges, streak tracking, and experience points to increase student motivation and engagement.
- **Proctoring Features:** Implement browser tab-switch detection and webcam proctoring capabilities for formal examination use cases requiring invigilation.
- **Analytics for Administrators:** Provide institution-level analytics showing class-wide performance trends, most-missed questions, and subject difficulty indices to support data-driven curriculum decisions.

Certificates





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This professional has demonstrated initiative and a
commitment to deepening their skills and advancing
their career. Well done!

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Krishna Kumar
CEO, Simplilearn

